



Efficient economic operation based on load dispatch of power systems using a leader white shark optimization algorithm

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Abstract

This article proposes the use of a leader white shark optimizer (LWSO) with the aim of improving the exploitation of the conventional white shark optimizer (WSO) and solving the economic operation-based load dispatch (ELD) problem. The ELD problem is a crucial aspect of power system operation, involving the allocation of power generation resources to meet the demand while minimizing operational costs. The proposed approach aims to enhance the performance and efficiency of the WSO by introducing a leadership mechanism within the optimization process, which aids in more effectively navigating the complex ELD solution space. The LWSO achieves increased exploitation by utilizing a leader-based mutation selection throughout each generation of white sharks. The efficacy of the proposed algorithm is tested on 13 engineer benchmarks non-convex optimization problems from CEC 2020 and compared with recent metaheuristic algorithms such as dung beetle optimizer (DBO), conventional WSO, fox optimizer (FOX), and moth-flame optimization (MFO) algorithms. The LWSO is also used to address the ELD problem in different case studies (6 units, 10 units, 11 units, and 40 units), with 20 separate runs using the proposed LWSO and other competitive algorithms being statistically assessed to demonstrate its effectiveness. The results show that the LWSO outperforms other metaheuristic algorithms, achieving the best solution for the benchmarks and the minimum fuel cost for the ELD problem. Additionally, statistical tests are conducted to validate the competitiveness of the LWSO algorithm.

Keywords Economic load dispatch · White shark optimizer · Optimization algorithms · Leader strategy

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1 Introduction

As is widely recognized, the extensive and reckless exploitation and allocation of power and other energy sources has led to limited and expensive resources, resulting in significant resource wastage and gradual impacts on social life. Therefore, it is imperative to make efficient and rational use of these limited energy resources. Utilizing energy and fostering innovation in energy-related areas have always been crucial factors in economic development and ensuring livelihood security. In particular, the rational and effective distribution of large-scale power resources is of utmost significance [1]. Economic load dispatch (ELD) is a crucial matter for power system operations. The ELD aims to reduce fuel costs while fulfilling inequality and equality constraints. Equality constraints are based on the stability of power between the productions of the system and the whole demand with transmission losses. Inequality constraints are based on power output [2]. Many traditional approaches have been proposed to solve ELD problems as linear programming, nonlinear programming, and mixed integer linear programming [3]. Nevertheless, the real-world system presents several significant considerations, including factors such as the valve-point effect, transmission losses, and the inherent nonlinearity of the generator set. ELD scheduling involves non-convex optimization challenges, making it exceedingly challenging to address these nonlinear problems using conventional mathematical approaches. Recently, metaheuristic techniques are widely used to solve the ELD problem [4]. In both unimodal and multimodal problems, exploration and exploitation can be accomplished using mathematical equations, which are implemented using specific techniques. However, the requirements for these techniques may differ depending on the problem at hand.

Metaheuristic algorithms are employed to solve various optimization problems [5–7]. The earliest algorithm used to solve the ELD problem is a genetic algorithm (GA) and its variants [8], particle swarm optimization (PSO) technique its variants [9], an improved slime mold algorithm (ISMA) algorithm [10], and artificial cooperative search (ACS) algorithm in [11].

Recently, numerous metaheuristic algorithms have been established to solve the ELD problem, such as the evolutionary simplex adaptive Hooke–Jeeves algorithm [12], opposition-mutual learning differential evolution with hybrid mutation strategy [13], hybrid gray wolf optimizer (HGWO) [14], improved fitness-dependent optimizer (IFDO) [15], a multigroup marine predator algorithm [16], chaotic eagle-strategy supply–demand optimizer (CESSDO) [17], a chaotic slime mold algorithm (CSMA) [18], and elementary function disturbance arithmetic

optimization technique (EFDAO) [19]. In 2018, Singh et al. suggested an enhanced gray wolf optimizer (GWO) to solve the ELD problem in collaboration with exploration and exploitation, which coordinates the behavior of gray wolf, random search, local random search, and reverse learning heuristic [20]. In [21], the sine–cosine algorithm was applied to solve the ELD problem. A comparative analysis to assess the behaviors of hurricane and sine-cosine optimizers was carried out for solving the ELD with considering the ecological emissions [22]. The enhanced moth-flame optimization algorithm was presented in [23] to solve different sizes of ELD problem considering the emission minimization. The enhanced social network search (ESNS) was suggested to solve the ELD problem using high- and low-velocity ratio approach to improve the searching balance between exploration and exploitation of each solution in [24]. In 2020, Ling et al. proposed the shrinking Gaussian distribution quantum performance optimization algorithm (SG-QPSO) to solve the ELD problem. By iteratively reducing the Gaussian probability distribution near the learning tendency point of each particle, SG-QPSO maintains a strong global search ability at the beginning and gradually enhances the local search ability [25]. To overcome the ELD difficulties, an enhanced arithmetic optimization is suggested. There are two crucial variables in the arithmetic optimization algorithm (AOA) math optimizer acceleration and probability [26]. It can be seen from the above literature that intelligent optimization algorithms were used to solve the ELD problem. Some directly use the original algorithm to solve the ELD problem, and some use some strategies to optimize the original algorithm to enhance its performance. It achieved better results in solving the ELD problem. A comparative study is shown in Table 1 to keep track of algorithms and their evaluation on standard test systems.

Recently, the white shark optimizer (WSO) has been suggested to emulate the manners of white sharks, containing their unique intelligences of audible range and smell when navigating and hunting [46]. In this article, an effective WSO algorithm is employed to achieve the optimal solution for the ELD problem. The main reason for choosing this algorithm in this article is that it has a few adjustable parameters that have been implemented easily. This characteristic makes it very potential for applications in many engineering fields. In [47], the WSO approach is applied in OPF solution of power systems with renewable energy sources. The WSO technique is suffering from slowing the convergence rate and falling in the local optima because of unbalance between the exploration and exploitation process. Therefore, the traditional WSO algorithm needs to be further improved [48]. Thus, the authors in this work aim to achieve a fine balance between them using the leader-based mutation-selection approach to

Table 1 Comparative study of several algorithms and their evaluation on standard test systems

Methodology	Test systems							
	3 units	6 units	10 units	11 units	13 units	15 units	20 units	40 units
Biogeography-based optimization (BBO) [27]			✓				✓	✓
Cultural self-organizing migrating algorithm (CSOMA) [28]					✓			✓
Adaptive real coded genetic algorithm (ARCGA) [29]			✓					✓
Modified differential evolution (MDE)[30]					✓	✓		✓
Artificial bee colony algorithm (ABC) [31]			✓		✓	✓		✓
Firefly algorithm (FA) [32]					✓	✓		✓
Novel direct search method (NDS) [33]	✓				✓			✓
Novel stochastic search method (NSS) [34]	✓				✓			✓
Seeker optimization algorithm (SOA) [35]								✓
Group search optimizer (GSO)[36]							✓	✓
Sensing cloud optimization (SCO) [37]						✓		✓
Shuffled differential evolution (SDE) [38]	✓				✓			✓
Real coded chemical reaction optimization (RCCRO) [39]						✓		
Chaotic bat algorithm (CBA) [40]					✓		✓	✓
Hybrid gray wolf optimizer (HGWO) [41]		✓				✓		✓
Improved adaptive differential evolution optimizer (IADE) [42]		✓						✓
Equilibrium optimizer (EO) [43]					✓	✓		✓
Clustering cuckoo search optimization (CCSO) [44]		✓	✓		✓	✓		✓
Hybrid salp swarm algorithm (HSSA) [2]	✓				✓			✓
Chaotic slime mold algorithm (CSMA) [18]		✓			✓	✓		✓
Enhanced beluga whale optimization (EBWO) [45]				✓				✓
Improved slime mold algorithm (ISMA) [10]		✓	✓	✓				✓

get more promising results than those gained from the conventional WSO algorithm, and it can improve search accuracy and extend global search capabilities.

The ELD problem has received significant attention in recent decades, as highlighted in the summary above. Although various optimization approaches have been proposed to tackle ELD problems. This article specifically focuses on integrating algorithm and constraint processing technology to address the ELD problems. In the context of constraint optimization problems, constraint-handling techniques are just as crucial as algorithm design, as such problems require managing both fitness function and the degree of constraint violation. Building on the considerations, a leader white shark optimizer (LWSO) is presented to solve the ELD problem. By incorporating a leader-based mutation-selection strategy, the LWSO algorithm aims to not only optimize the objective function but also effectively manage the constraints involved in the problem. This leader-based approach enables the algorithm to adapt to changing dynamics in the optimization landscape and ensures that the solution obtained adheres to the problem's

constraints, enhancing the overall robustness of the ELD optimization process. Furthermore, the LWSO algorithm provides a promising avenue for further advancements in solving complex optimization problems, by striking a balance between exploration and exploitation to achieve more accurate and globally competitive solutions. The vital contributions of the article could be brief as below:

- Applying the proposed LWSO to solve the ELD problem;
- Four test networks with 6-, 10-, 11-, and 40-hermal units are used to confirm the LWSO performance;
- Comparison between the LWSO and other recent algorithms such as the Golden Jackal Optimization (GJO), Northern Goshawk Optimization (NGO), fox optimizer (FOX), and conventional WSO algorithms and other literature algorithms such as the Improved slime mold algorithm (ISMA), tunicate swarm algorithm (TSA), Harris Hawks optimizer (HHO), slime mold algorithm (SMA), jellyfish search optimizer (JS), and PSO algorithms is performed.

- Statistical analysis is performed for 20 trails of studied techniques, and the strength and convergence rates for LWSO are discussed.

The rest of the paper is arranged as Sect. 2 presents the ELD problem's mathematical model; the original WSO and the proposed LWSO techniques are explained in Sect. 3; the results and discussions are shown in Sect. 4; and finally, Sect. 5 presents conclusion of the article.

2 Mathematical model of ELD problem

The fuel cost function for this problem follows a quadratic form, and the total fuel costs take into account the valve-point effects in accordance with the output of thermal generations while satisfying their respective constraints. The function can be expressed as follows [49]:

$$F_1 = \sum_{i=1}^{N_G} [a_i + b_i P_{Gi} + c_i P_{Gi}^2 + |d_i \sin(e_i (P_{Gi}^{min} - P_{Gi}))|] \quad (1)$$

where a_i , b_i , and c_i are the cost coefficients for the i_{th} unit; P_i refers to the power output of these units; and N_G denotes the number of generations.

The balance of power constraint in this problem ensures that the total output power generation P_T (in MW) is equal to the sum of the total load demand P_D (in MW) and the power loss P_{Loss} (in MW) in the entire system. This equality constraint can be mathematically represented as follows [50]:

$$P_T = \sum_{i=1}^{N_G} P_{Gi} = P_D + P_{Loss}, \quad (2)$$

P_{Loss} denotes the active output power of the units, and it can be calculated from Kron's loss formula as below:

$$P_{Loss} = \sum_{i=1}^{N_G} \sum_{j=1}^{N_G} P_{Gi} B_{ij} P_{Gj} + \sum_{i=1}^{N_G} B_{0i} P_{Gi} + B_{00}, \quad (3)$$

where B_{00} , B_{0i} , and B_{ij} refer to the coefficients of the power loss.

The inequality bound is attained during each thermal generator works within its operating constraints as follows:

$$P_{Gi}^{min} \leq P_{Gi} \leq P_{Gi}^{max}, \quad (4)$$

where P_{Gi}^{min} , and P_{Gi}^{max} represent the operating bounds of generator i .

The constraint violation can be expressed as follows [51]:

$$V = \left| \sum_{i=1}^{N_G} P_{Gi} - P_D - P_{Loss} \right| \quad (5)$$

3 Methodology

3.1 Conventional WSO

The WSO algorithm is inspired by the behaviors of the white sharks when hunting to assist them to live in the ocean depths [46]. Its results prove its strength in solving different types of optimization problems. Therefore, the WSO provides numerous advantages for composite optimization problems. The models of the WSO algorithm determine the optimal value of the fuel cost. This section briefly describes the basics of the proposed WSO algorithm.

3.1.1 Initialization of WSO

The initialization position of each white shark can be calculated from the following equation [47]:

$$w = \begin{bmatrix} w_1^1 & w_2^1 & \dots & w_d^1 \\ w_1^2 & w_2^2 & \dots & w_d^2 \\ \dots & \dots & \dots & \dots \\ w_1^n & w_2^n & \dots & w_d^n \end{bmatrix} \quad (6)$$

In this context, the variable w represents the position of the white sharks in the search area, while the variable d represents the number of variables in the problem.

3.1.2 Update the parameters of the WSO algorithm

$$v = [n \times \text{rand}(1, n)] + 1 \quad (7.1)$$

$$p_1 = p_{max} + (p_{max} - p_{min}) \times e^{-(4 \times \frac{v}{K})^2} \quad (7.2)$$

$$p_2 = p_{min} + (p_{max} - p_{min}) \times e^{-(4 \times \frac{v}{K})^2} \quad (7.3)$$

$$\mu = \frac{2}{\left| 2 - \tau - \sqrt{\tau^2 - 4\tau} \right|} \quad (7.4)$$

$$a = \text{sgn}(w_k^j - u) > 0 \quad (7.5)$$

$$b = \text{sgn}(w_k^j - u) < 0 \quad (7.6)$$

$$w_0 = \oplus(a, b) \quad (7.7)$$

$$f = f_{min} + \frac{f_{max} - f_{min}}{f_{max} + f_{min}} \quad (7.8)$$

$$mv = \frac{1}{a_0 + e^{(\frac{K}{2} - K)/a_1}} \quad (7.9)$$

$$S_S = \left| 1 - e^{(-a_2 \times \frac{K}{2})} \right| \quad (7.10)$$

where v refers to the white sharks' index vector getting the optimal location defined p_1 and p_2 denote the white sharks' forces, and μ denotes the constriction factor proposed in WSO. a and b refer to one-dimensional binary vectors w_0

denotes a logical vector. f represents the frequency of the white shark's wavy motion. mv denotes the movement force which is related to the number of iterations. s_s denotes a parameter proposed to rapid the sense's strength of smell and sight of the sharks during following other white sharks that are close to optimum prey.

3.1.3 Speed of movement to prey

When a white shark identifies a prey's position using hearing a pause in the waves because the prey moves, as below:

$$u_{k+1}^i = \mu \left[u_k^i + p_1 (w_{gbest_k} - w_k^i) \times c_1 + p_2 (w_{best}^i - w_k^i) \times c_2 \right] \quad (8)$$

the new rapidity vector of the i^{th} shark can be represented by v_{k+1}^i .

3.1.4 Movement in the best prey's way

In this scenario, the movement of sharks as their approached fish was depicted.

$$w_{k+1}^i = \begin{cases} w_k^i \cdot \rightarrow \oplus w_0 + u \cdot a + l \cdot b; & rand < mv \\ w_k^i + u_k^i / f; & rand \geq mv \end{cases} \quad (9)$$

3.1.5 Movement in the way of the best shark

Equation (10) expresses how the white sharks can maintain their position ahead of the most beneficial one that is closer to the goal. This phenomenon is achieved through the following equation:

$$w_{k=1}^i = w_{gbest_k} + r_1 \vec{D}_w \text{sgn}(r_2 - 0.5) \quad r_3 < S_s \quad (10)$$

w_{k+1}^i is the upgraded shark's location, $\text{sgn}(r_2 - 0.5)$ returns 1 or -1 to adapt the search track, and r_1 , r_2 and r_3 are the random values. D_w denotes the length for both goal and shark, and it can be calculated as follows:

$$\vec{D}_w = |rand \times (w_{gbest} - w_k^i)| \quad (11)$$

The flowchart of the WSO algorithm is shown in Fig. 1.

3.2 The procedure of the improved LWSO algorithm

The proposed method to address the possibility of the optimum value dropping into local minima is called leader-based mutation selection [52]. This modification relies on the best position vector (w_{best-1}^{iter}), the second-best position vector (w_{best-1}^{iter}), and the third-best position vector (x_{best-2}^t)

based on their objective function values for the new position vector ($w_i(new)$) between the members of the population. The new mutation position vector ($w_i(mut)$) is calculated using the following equation:

$$w_i(mut) = w_i(new) + 2 \times \left(1 - \frac{iter}{Max_iters} \right) \times (2 \times rand - 1) (2 \times w_{best}^{iter} - (w_{best-1}^{iter} + w_{best-2}^{iter})) + (2 \times rand - 1) (w_{best}^{iter} - w_i(new)) \quad (12)$$

After that, the following location is reorganized as follows:

$$w_i(iter + 1) = \begin{cases} w_i(mut) f(w_i(mut)) < f(w_i(new)) \\ w_i(new) f(w_i(new)) \geq f(w_i(new)) \end{cases} \quad (13)$$

Lastly, the optimal solution is updated by the following equation:

$$w_{best} = \begin{cases} w_i(mut) f(w_i(mut)) < f(w_{best}) \\ w_i(new) f(w_i(new)) < f(w_{best}) \end{cases} \quad (14)$$

Figure 2 displays the flowchart of the LWSO algorithm, which includes the residence of the leader-based mutation selection. Moreover, Algorithm 1 describes the LWSO algorithm's pseudocode. This improvement enhances the exploration of the LWSO technique through simultaneous crossover and mutation using the three best leaders. The combination of crossover and mutation in genetic algorithms is a powerful strategy to address trapping and slow convergence. Crossover facilitates the sharing of valuable genetic information, while mutation introduces randomness and diversity, collectively promoting a more effective and efficient exploration–exploitation trade-off. The proposed LWSO algorithm offers several advantages over the original WSO technique:

The advantages of LWSO can be concluded as follows:

1. **Enhanced Exploration:** Simultaneous crossover and mutation using the three best leaders can lead to increased diversity in the search space. This enhancement allows the algorithm to explore a wider range of solutions, potentially leading to better optimization outcomes.
2. **Improved Convergence:** Combining crossover and mutation can help the algorithm converge faster to better solutions. Crossover promotes the sharing of genetic information between solutions, while mutation introduces random variations that can help escape local optima.
3. **Better Solution Quality:** By leveraging the best leaders, the algorithm can focus on the most promising regions of the search space, increasing the likelihood of finding high-quality solutions.

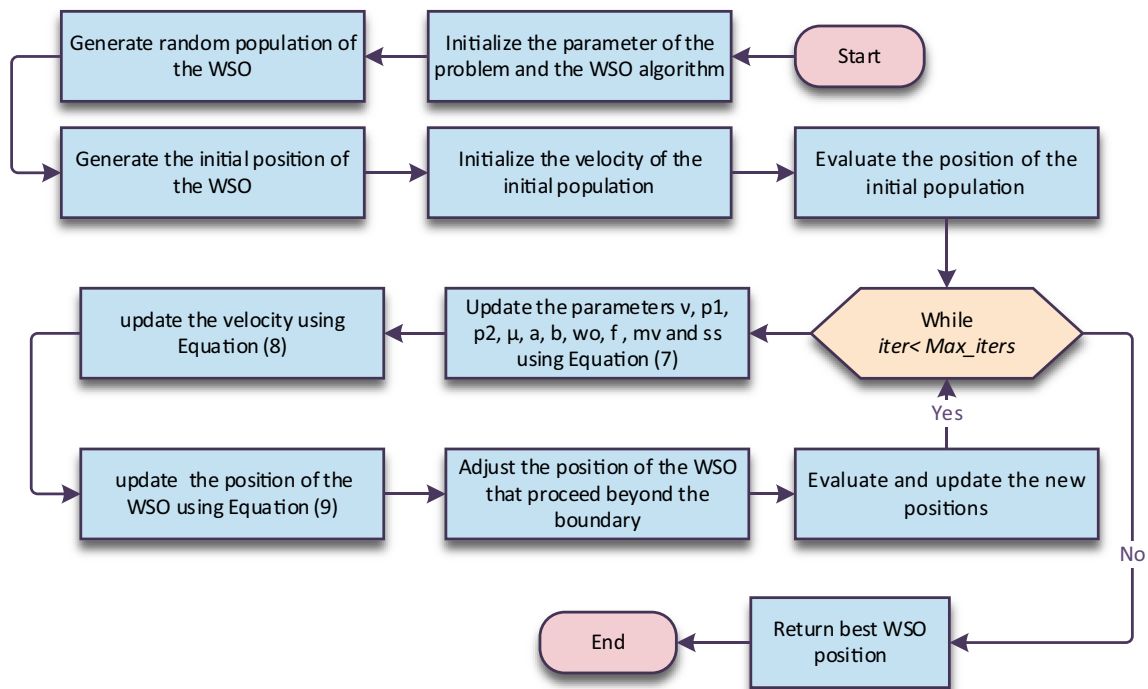


Fig. 1 The flowchart of the proposed WSO algorithm

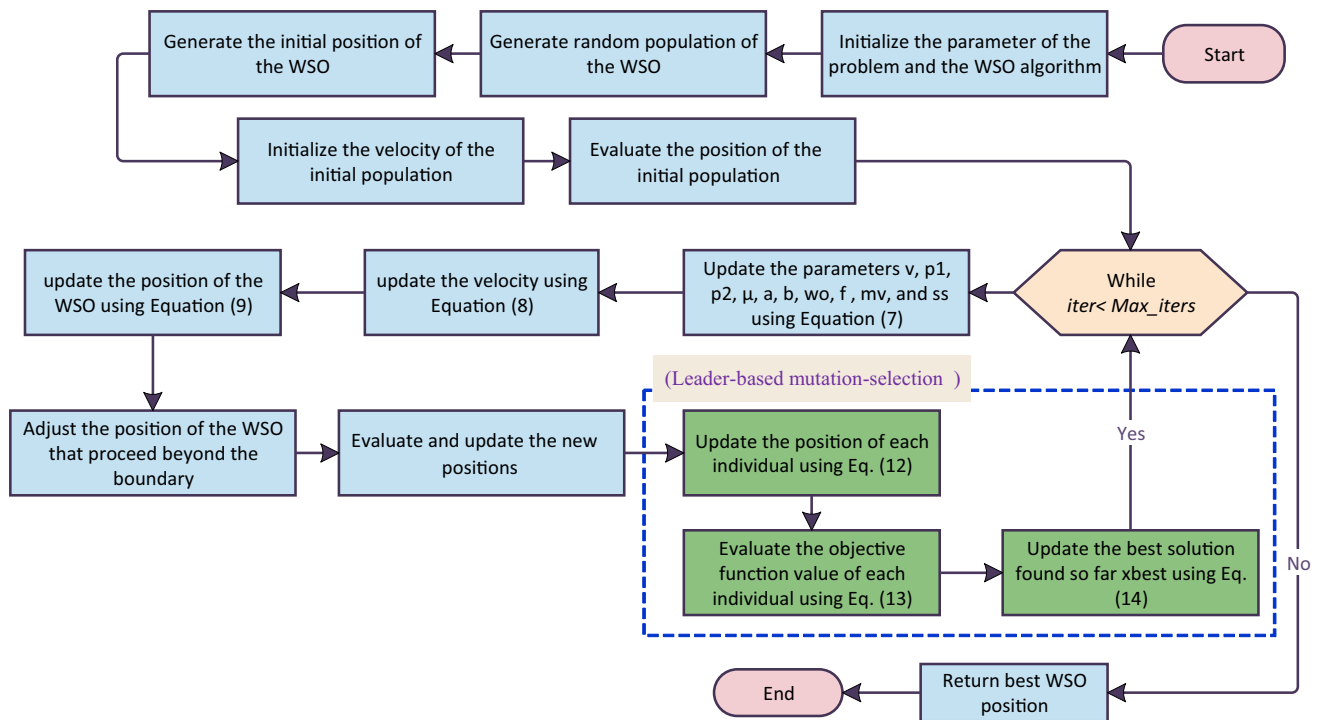


Fig. 2 Flowchart of proposed LO algorithm

However, like any algorithm, LWSO may have its own shortcomings. One potential disadvantage of the proposed LWSO algorithm could be:

Increased Computational Complexity: Simultaneous crossover and mutation can significantly increase the computational requirements of the algorithm. This can lead to longer execution times and higher resource usage, making it less suitable for problems with strict time or resource constraints.

Algorithm 1: Pseudocode of the LWSO algorithm

3.3 Analysis of algorithm computational complexity

The time complexity of the proposed LWSO algorithm depends on the population size (N), the maximum number of iterations (T), the dimensions of the problem (D), and the function assessment's cost (C). In the original WSO, the time complexity of initializing the population is $O(N \times D)$. The time complexity of the assessment of the cost function demands is $O(N \times C \times T)$. The time complexity of updating positions is $O(N \times D \times T)$. Therefore, the general

Set the control parameter (Dimension of problem (d), maximum iteration, and population size).

Initialize the population randomly

Initialize the velocity of the initial population

Evaluate the fitness of the new solution

Obtain the best solution

While $k \leq K$

Update the parameters v , p_1 , p_2 , μ , a , b , f , mv , and S_s

For $i=1:n$

Update the velocity using Eq. (8)

End For

For $i=1:n$

If $\text{rand} < mv$

Update the position using Eq. (9-1)

Else

Update the position using Eq. (9-2)

End If

End For

For $i=1:n$

Update the position using Eq. (10)

End For

Check the limits of the new locations and evaluate the fitness values

%Leader-based- mutation selection

Update the new location using Eq. (12)

Check the limits of the new locations and evaluate the fitness values

Find the new solution if the fitness is better

$k=k+1$

End while

Output the best solution

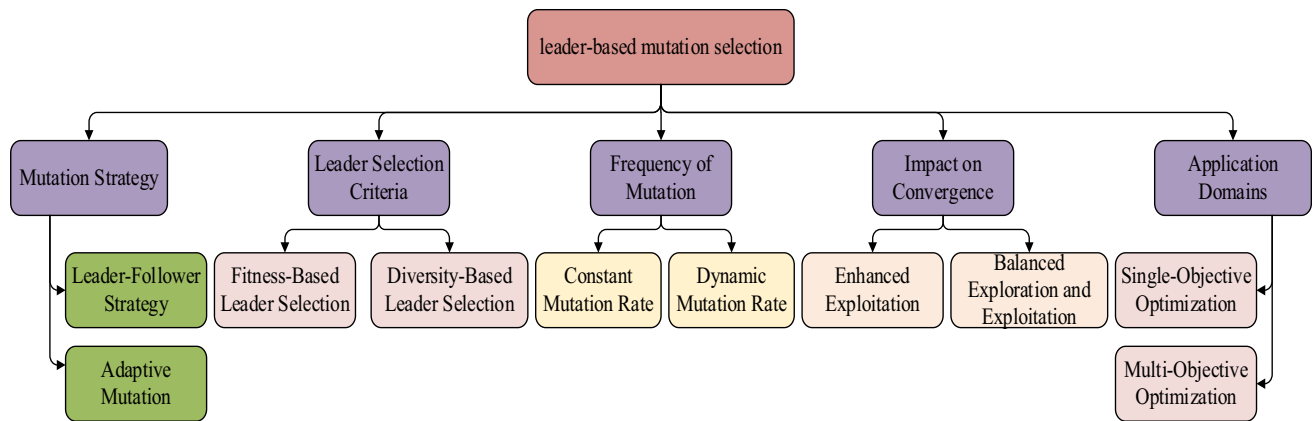


Fig. 3 The taxonomy of leader-based mutation selection

Table 2 Real-world optimization engineering benchmark cases included in the CEC 2020 and their bounds [53]

Function	Case study and its lower and upper bounds	Decision variables	Constraints	Global optima
RC8	Process synthesis problem: $0 \leq x_1 \leq 1.6$; $x_2 \in \{0, 1\}$	2	2	2
RC12	Process synthesis problem: $0 \leq x_2, x_3$; $x_1 \leq 100$; $x_7, x_6, x_5, x_4 \in \{0, 1\}$	7	9	2.92
RC13	Process design problem: $27 \leq x_3$; $x_1 \leq 1$; $x_2 \leq 45$; $x_4 \in \{78, 79, \dots, 102\}$; $x_5 \in \{33, 34, \dots, 45\}$	5	3	26,900
RC15	Weight minimization of a speed reducer $0.7 \leq x_2 \leq 0.8$; $17 \leq x_3 \leq 28$; $2.6 \leq x_1 \leq 3.6$; $5 \leq x_7 \leq 5.5$; $7.3 \leq x_5$; $x_4 \leq 8.3$; $2.9 \leq x_6 \leq 3.9$	7	11	2990
RC17	Tension/compression spring design (case 1) $0.05 \leq x_1 \leq 2.00$; $0.25 \leq x_2 \leq 1.30$; $2.00 \leq x_3 \leq 15.0$	3	3	0.0127
RC18	Pressure vessel design with integer variables $10 \leq x_1 \leq 4$; $x_3 \leq 200$; $1 \leq x_2$; $x_1 \leq 99$	4	4	5890
RC19	Welded beam design: $0.1 \leq x_3$; $x_2 \leq 10$; $0.1 \leq x_4 \leq 2$; $0.125 \leq x_1 \leq 2$	4	5	1.67
RC20	Three-bar truss design problem: $0 \leq x_1$; $x_2 \leq 1$	2	3	264
RC21	Multiple disk clutch brake design problem: $60 \leq x_1 \leq 80$; $90 \leq x_2 \leq 110$; $1 \leq x_3 \leq 3$; $0 \leq x_4 \leq 1000$; $2 \leq x_5 \leq 9$	5	7	0.0235
RC28	Rolling element bearing: $x_1 \in \{125, 150\}$; $x_2 \in \{10.5, 31\}$; $x_3 \in \{4.51, 50.49\}$; $x_4 \in \{0.515, 0.6\}$; $x_5 \in \{0.515, 0.6\}$; $x_6 \in \{0.4, 0.5\}$; $x_7 \in \{0.6, 0.7\}$; $x_8 \in \{0.3, 0.4\}$; $x_9 \in \{0.02, 1\}$; $x_{10} \in \{0.6, 0.85\}$	10	9	14,600
RC29	Gas transmission compressor design (GTCD); $20 \leq x_1 \leq 50$; $1 \leq x_2 \leq 10$; $20 \leq x_3 \leq 50$; $0.1 \leq x_4 \leq 60$	4	1	2,960,000
RC31	Gear train design problem: $12 \leq x_2$; $x_3, x_4, x_1 \leq 60$	4	1	0
RC32	Himmel Lau's function: $78 \leq x_1 \leq 102$; $33 \leq x_2 \leq 45$; $27 \leq x_3, x_4, x_5 \leq 45$;	5	6	-30,700

time complexity of WSO is $O(N \times D + N \times C \times D + N \times D \times T)$. The time complexity of LWSO is $O(2 \times N \times D \times T + N \times D + N \times D \times T)$.

3.4 Taxonomy of leader-based mutation selection

Forming a taxonomy for leader-based mutation selection includes categorizing and organizing the key elements

related to this specific optimization technique. Figure 3 shows a taxonomy for leader-based mutation selection.

4 Simulation results and discussion

4.1 Simulation results of real-world engineering problems

To evaluate the performance of the LWSO algorithm in solving the real-world non-convex, constrained

Table 3 Statistical indices for the engineering benchmark studied cases

Functions	Index	WSO	LWSO	DBO	FOX	MFO
RC8	Min	2	2	2	2	2
	Average	2	2	2	2	2
	Median	2	2	2	2	2
	Max	2	2	2	2	2
	Std	2.88E-16	2.44E-16	2.44E-16	7.1E-08	2.28E-16
	Rank	4	1	1	5	1
RC12	Min	2.924831	2.924831	2.924831	2.924872	2.924839
	Average	2.924831	2.940521	3.378957	2.946508	2.96061
	Median	2.924831	2.924831	3.081732	2.925031	2.946961
	Max	2.924831	3.081732	4.074353	3.082564	3.081732
	Std	1.43E-07	0.048293	0.481632	0.047529	0.053133
	Rank	1	2	5	3	4
RC13	Min	26,887.42	26,887.42	26,887.42	26,887.42	26,887.42
	Average	26,887.42	26,887.42	26,887.42	27,135.01	26,887.42
	Median	26,887.42	26,887.42	26,887.42	26,887.42	26,887.42
	Max	26,887.42	26,887.42	26,887.42	28,368.22	26,887.42
	Std	8.52E-05	2.05E-06	1.12E-11	473.4105	1.12E-11
	Rank	4	3	1	5	1
RC15	Min	2994.648	2994.429	2994.424	2995.595	2994.424
	Average	6E + 14	5E + 13	6E + 14	6.5E + 14	2998.352
	Median	1E + 15	2994.439	1E + 15	1E + 15	2994.424
	Max	1E + 15	1E + 15	1E + 15	1E + 15	3033.702
	Std	5.03E + 14	2.24E + 14	5.03E + 14	4.89E + 14	12.08925
	Rank	3	2	4	5	1
RC17	Min	0.012665	0.012665	0.012666	0.012677	0.012674
	Average	0.012665	0.012665	0.012742	5E + 13	0.012827
	Median	0.012665	0.012665	0.012719	0.012781	0.012719
	Max	0.012665	0.012665	0.012928	1E + 15	0.014283
	Std	1.69E-08	8.3E-13	7.56E-05	2.24E + 14	0.000356
	Rank	2	1	3	5	4
RC18	Min	6247.675	6247.72	6247.673	6359.528	6247.673
	Average	6247.681	6247.934	6544.502	39,927.73	6283.05
	Median	6247.681	6247.853	6382.985	15,046.06	6247.673
	Max	6247.688	6248.808	7319.001	239,304.1	6436.743
	Std	0.003228	0.236775	400.6724	65,645.18	63.7404
	Rank	1	2	4	5	3
RC19	Min	1.670218	1.670218	1.670218	1.67593	1.670218
	Average	1.670218	1.670218	1.700254	1.756922	1.670219
	Median	1.670218	1.670218	1.670218	1.722726	1.670218
	Max	1.670218	1.670218	1.816712	1.994586	1.670239
	Std	6.2E-08	5.61E-08	0.055884	0.080852	4.83E-06
	Rank	2	1	4	5	3
RC20	Min	263.8958	263.8958	263.8958	263.8958	263.8958
	Average	263.8958	263.8958	263.8959	263.8959	263.8985
	Median	263.8958	263.8958	263.8959	263.8959	263.8967
	Max	263.8958	263.8958	263.8961	263.8962	263.9237
	Std	4.29E-12	1.3E-14	4.76E-05	7.87E-05	0.006081
	Rank	2	1	3	4	5

Table 3 (continued)

Functions	Index	WSO	LWSO	DBO	FOX	MFO
RC21	Min	0.235242	0.235242	0.235242	0.235242	0.235242
	Average	0.235242	0.235242	0.235242	0.235243	0.235242
	Median	0.235242	0.235242	0.235242	0.235243	0.235242
	Max	0.235242	0.235242	0.235242	0.235243	0.235242
	Std	5.94E-09	1.87E-11	1.14E-16	9.23E-08	1.14E-16
	Rank	4	3	1	5	1
RC28	Min	5599.448	5599.448	5599.448	5599.448	5599.448
	Average	5599.448	5599.448	5599.448	5599.448	5599.448
	Median	5599.448	5599.448	5599.448	5599.448	5599.448
	Max	5599.448	5599.448	5599.448	5599.448	5599.448
	Std	0	0	0	0	0
	Rank	1	1	1	1	1
RC29	Min	2,964,895	2,964,895	2,964,895	2,989,723	2,964,897
	Average	2,964,895	2,964,895	3,011,451	3,086,914	2,965,099
	Median	2,964,895	2,964,895	2,964,897	3,096,913	2,964,998
	Max	2,964,895	2,964,895	3,147,942	3,104,538	2,966,063
	Std	9.68E-05	1.85E-05	74,628.94	29,668.78	302.1104
	Rank	2	1	4	5	3
RC31	Min	3.89E-20	5.46E-19	0	3.38E-19	0
	Average	3.91E-16	8.34E-17	0	4.3E-17	0
	Median	7.25E-17	1.44E-17	0	6.21E-18	0
	Max	1.95E-15	4.52E-16	0	3.28E-16	0
	Std	5.92E-16	1.33E-16	0	8.09E-17	0
	Rank	5	4	1	3	1
RC32	Min	−30,665.5	−30,665.5	−30,665.5	−30,665.5	−30,665.5
	Average	−30,665.5	−30,665.5	−30,665.5	−30,648.6	−30,665.5
	Median	−30,665.5	−30,665.5	−30,665.5	−30,665.5	−30,665.5
	Max	−30,665.5	−30,665.5	−30,665.5	−30,514.8	−30,665.5
	Std	0.001059	0.002432	3.73E-12	41.5896	3.73E-12
	Rank	3	4	1	5	1
Average rank	2.615385	2	2.538462	4.307692	2.230769	
Final ranking	4	1	3	5	2	

engineering optimization problems, it is conducted tests on 13 optimization problems in chemical and mechanical engineering, sourced from CEC 2020. The constraint functions' lower and upper limits violation were obtained from [53], and the obtained results were compared with other optimization solvers including DBO [54], WSO, FOX [55], and MFO [56]. Table 2 provides essential information about the benchmark functions used in the study. As all these problems have multiple inequality constraints, any algorithm designed to solve them must integrate a constraint-handling technique. Common approaches include repairing, decoding, preserving, and penalizing, among others. In the presented case studies, the firm penalty method was implemented to handle the

constraints. The population is set as 200 for all 13 benchmarks. The maximum iterations equal 500 for the case studies.

Table 3 presents the results of statistical indices for CEC 2020 optimization problems, compared the performance of the LWSO algorithm and other competitive optimization algorithms and further indicates ranking for all the 13 case studies. From this table, the applied techniques are sorted. It can be seen from this ranking order that the LWSO algorithm superiors the other compared algorithms on 13 function problems. MFO and DBO display strong efficiency that are the second and third optimal. It can be concluded from this discussion that the LWSO algorithm is an effective algorithm for acquiring the optimal solutions of these non-convex constrained f problems. The

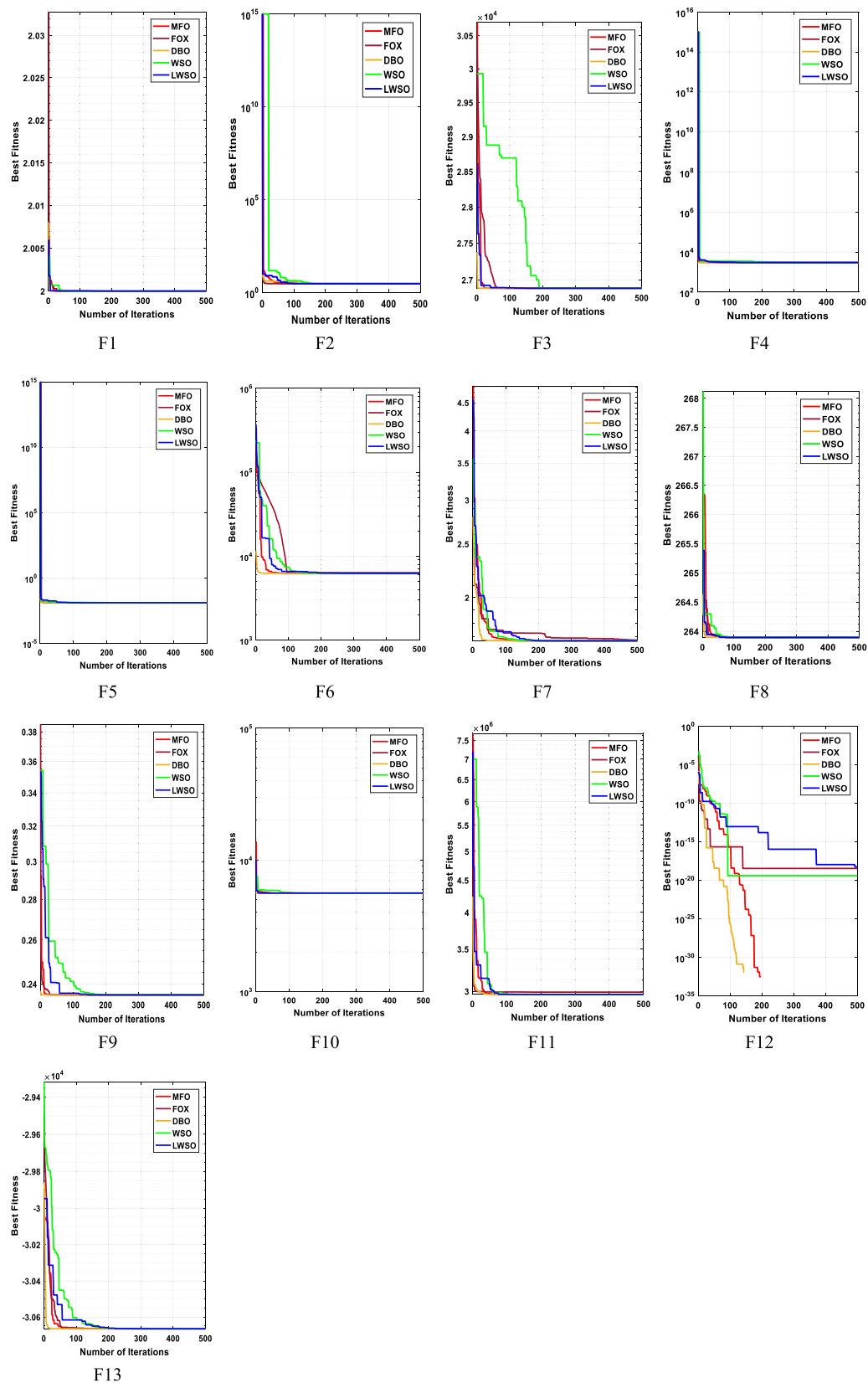


Fig. 4 The convergence curves for all techniques and benchmark functions

Fig. 5 Boxplots for all techniques and benchmark functions

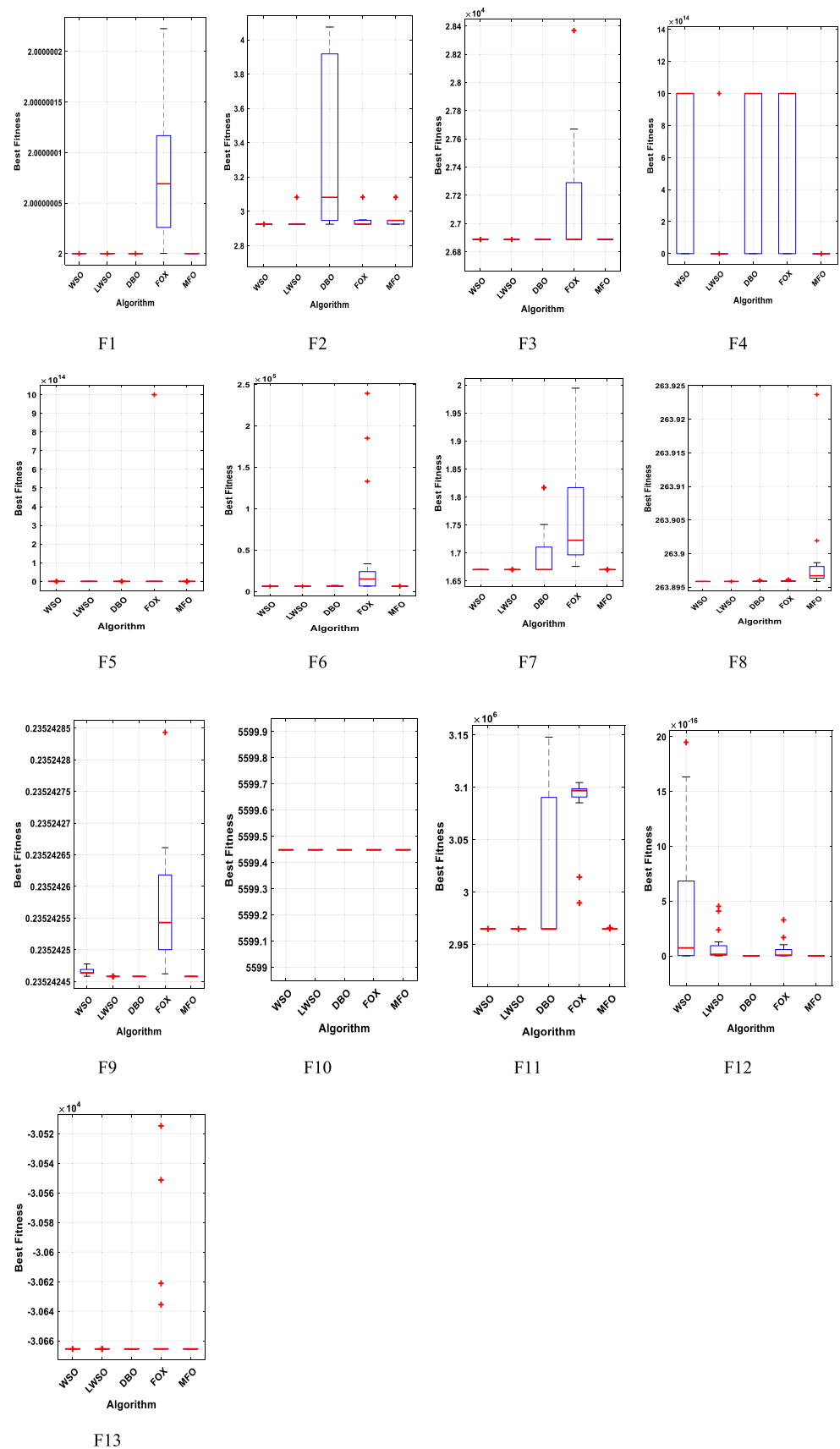


Table 4 Statistical results of the Wilcoxon rank-sum test

LWSO vs	WSO		DBO		FOX		MFO	
Function	<i>P</i>	Winner	<i>P</i>	Winner	<i>P</i>	Winner	<i>P</i>	Winner
F1	8.77E-03	+	1.00E + 00	=	1.13E-08	+	3.42E-01	=
F2	1.60E-05	+	1.96E-06	+	9.75E-06	+	1.30E-05	+
F3	6.01E-07	+	8.01E-09	−	6.80E-08	+	8.01E-09	−
F4	1.03E-06	+	1.56E-01	=	7.47E-07	+	3.50E-06	−
F5	6.80E-08	+	5.73E-08	+	6.80E-08	+	5.37E-08	+
F6	6.80E-08	−	1.60E-04	+	6.80E-08	+	2.83E-01	+
F7	1.20E-06	+	4.25E-01	=	6.80E-08	+	3.75E-04	+
F8	1.12E-08	+	1.13E-08	+	1.13E-08	+	1.13E-08	+
F9	6.80E-08	+	8.01E-09	−	6.80E-08	+	8.01E-09	−
F10	NaN	=	NaN	=	NaN	=	NaN	=
F11	1.43E-07	+	7.85E-08	+	6.80E-08	+	6.80E-08	+
F12	3.79E-01	=	8.01E-09	−	1.14E-01	=	8.01E-09	−
F13	1.20E-01	=	8.01E-09	−	9.17E-08	+	8.01E-09	−
WRST (+ / = / −)	9/3/1		5/4/4		11/2/0		6/2/5	

Table 5 Friedman test for the five algorithms

Function	LWSO	WSO	DBO	FOX	MFO
F1	2.35	3.05	2.35	5	2.25
F2	1.35	1.9	4.575	3.55	3.625
F3	3	4	1.5	5	1.5
F4	2.4	3.975	3.275	4.05	1.3
F5	1	2	3.825	4.5	3.675
F6	2.75	1.75	3.55	4.8	2.15
F7	2.3	3.45	2.75	4.85	1.65
F8	1	2	3.45	3.75	4.8
F9	3	4.1	1.5	4.9	1.5
F10	3	3	3	3	3
F11	1	2.05	3.55	4.8	3.6
F12	4.15	4.15	1.5	3.7	1.5
F13	3.4	3.6	1.5	5	1.5
Mean ranks	2.361538	3.001923	2.794231	4.376923	2.465385

Table 6 Parameter settings of the competitive algorithms and the proposed LWSO algorithm

Algorithm	Parameters
FOX	$C_1 = 0.18$; $C_2 = 0.82$
WSO	$fmin = 0.07$; $fmax = 0.75$; $\tau = 4.125$; $a0 = 6.25$; $a1 = 100$; $a2 = 0.0005$
LWSO	$fmin = 0.07$; $fmax = 0.75$; $\tau = 4.125$; $a0 = 6.25$; $a1 = 100$; $a2 = 0.0005$

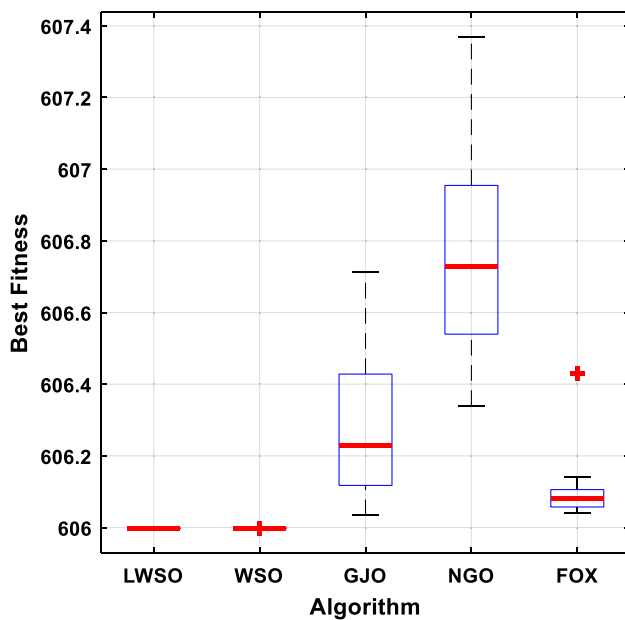
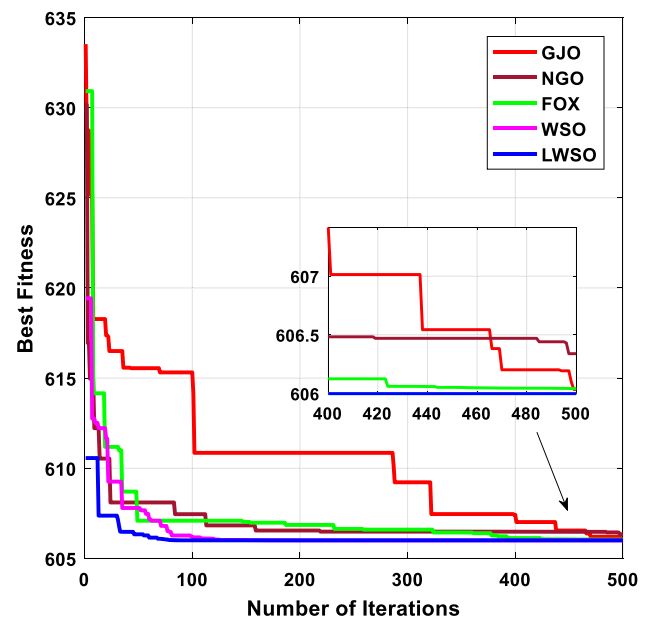
convergence rates of these algorithms for the tested functions are shown in Fig. 4. To approve the performance of the proposed algorithm, a boxplot of each algorithm and objective function is presented in Fig. 5. Figure 5 displays

the boxplots of the LWSO algorithm, for most of the functions, these plots are narrow and among the smallest values.

Table 7 Optimum solution values for the fuel cost of the first test system with six generators

Method	LWSO	WSO	GJO	NGO	FOX
P1 (MW)	0.120969	0.121006	0.107986	0.127545	0.122977
P2 (MW)	0.286312	0.286337	0.284068	0.255888	0.289468
P3(MW)	0.583557	0.583694	0.597299	0.638613	0.559545
P4 (MW)	0.992854	0.992772	0.996569	1.002144	0.994244
P5 (MW)	0.52397	0.523801	0.527366	0.485919	0.53871
P6 (MW)	0.351899	0.351949	0.346111	0.348028	0.355219
V (MW)	2.03E-11	2.56E-10	5.26E-08	8.42E-08	3.6E-07
PL	0.025562	0.025558	0.025399	0.024136	0.026164
Fuel cost (\$/h)	605.9984	605.9984	606.0342	606.3391	606.041

The best values obtained are in bold

**Fig. 6** Boxplots of various algorithms (system 1)**Fig. 7** Fuel cost convergence curves of various algorithms (system 1)

4.2 Wilcoxon's rank test results

In this subsection, the differences between the proposed LWSO and well-known optimization algorithms are further analyzed statistically using the Wilcoxon rank-sum test (WRST), which is a paired test that checks for significant differences between two algorithms. The results of the test between LWSO and each technique at a significance level of $\alpha = 0.05$ are presented in Table 4, where the symbols “+/-/-” show whether LWSO executes better, similarly, or worse than the comparison technique. This table also presents the statistical results of LWSO in different dimensions and functions, signifying whether LWSO performs better, similarly, or worse than the comparison algorithm. LWSO outperforms other comparative techniques in the statistics of 13 optimization problems in chemical and mechanical engineering, sourced from CEC

Table 8 Statistical results for fuel cost \$/h (case study 1)

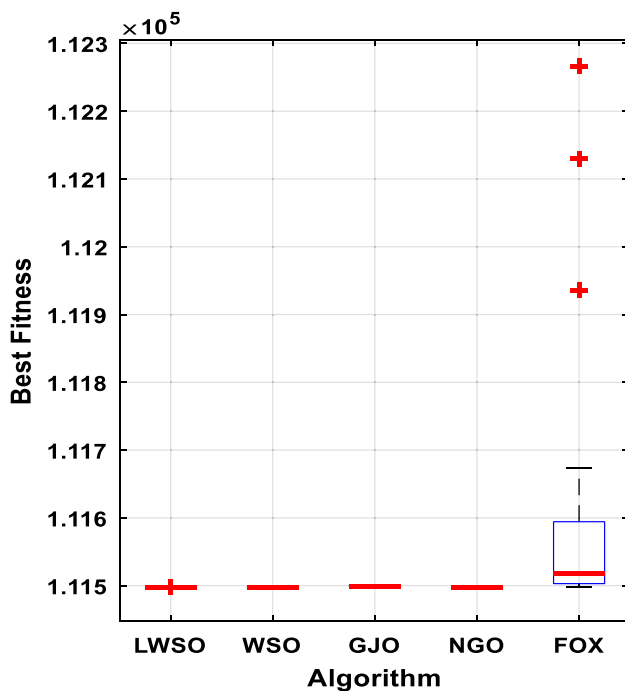
Method	Min	Max	Median	Average	Std
LWSO	605.9984	605.9984	605.9984	605.9984	2.33E-13
WSO	605.9984	605.9984	605.9984	605.9984	2.04E-07
GJO	606.0342	606.714	606.23	606.2743	0.186672
NGO	606.3391	607.369	606.7276	606.7602	0.285229
FOX	606.041	606.4303	606.0805	606.0977	0.082404
PSO	606.006	606.0501	606.0225	606.0253	0.0171
JS	606.3315	607.6245	606.9829	606.9632	0.323814
SMA	606.0017	606.1084	606.0159	606.0227	0.026
TSA	606.0248	606.4199	606.0501	606.1181	0.1508
HHO	606.5292	626.7676	611.3046	611.5524	3.826237
ISMA	605.9984	606.0374	606.0126	606.013	0.0142

The best values obtained are in bold

Table 9 Optimal solution of the fuel cost of the second test system with 10 generators

Method	LWSO	WSO	GJO	NGO	FOX
P1	55	54.99769	55	55	55
P2	80	79.98464	80	80	80
P3	106.9388	106.9542	107.5513	106.7959	107.1167
P4	100.5743	100.3228	100.1761	100.4317	100.1467
P5	81.5070	81.54307	81.88875	81.38856	80.10082
P6	83.0188	83.27021	82.40525	83.41902	84.6699
P7	300	299.9963	300	300	300
P8	340	339.9925	340	340	340
P9	470	469.9954	470	470	470
P10	470	469.9792	470	470	469.9999
V (MW)	1.47E-06	0.00114	0.021904	0.001767	7.21E-06
PL	87.03884	87.03719	87.04334	87.03696	87.03395
Fuel cost (\$/h)	111,497.63	111,497.76	111,498.46	111,497.71	111,498.25

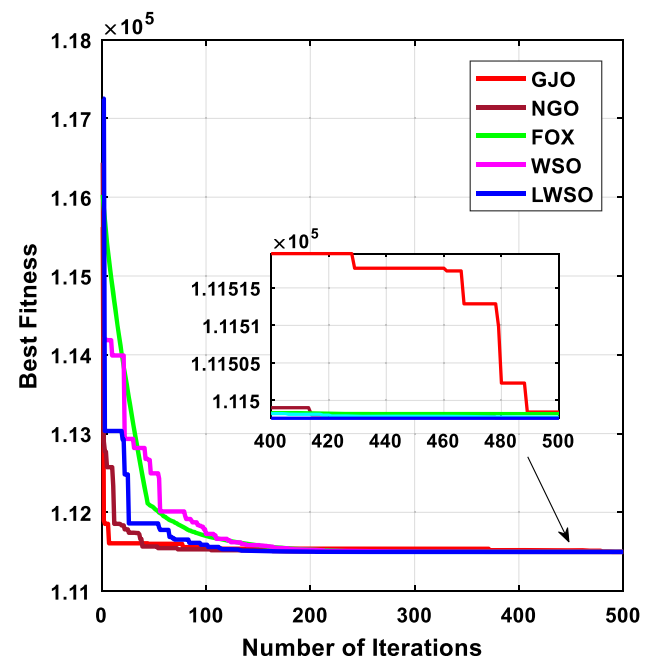
The best values obtained are in bold

**Fig. 8** Boxplots of various algorithms (System 2)

2020, which approves the significant dominance of LWSO in most functions compared to other techniques. Therefore, it can be concluded that the proposed LWSO technique exhibits the best performance compared to other algorithms.

4.3 Friedman's rank test results

Table 5 presents the statistical results obtained by Friedman tests [57]. The smaller the ranking value, the better the performance of the algorithm. From the results, we can get the ranks of five algorithms as follows: LWSO, MFO,

**Fig. 9** Fuel cost convergence curves of various algorithms (System 2)

DBO, WSO, and FOX. The highest-ranking shows that LWSO is the best algorithm among the five algorithms.

4.4 Results of the studied cases

The LWSO technique is investigated on several ELD problems. Also, a comparison is performed between the LWSO and other optimization algorithms including GJO [58], NGO [59], FOX, and conventional WSO algorithms. The coding and simulation of the LWSO technique are accomplished in the Matlab 2016a software. The LWSO technique and other well-known algorithms are run using a

Table 10 Statistical results for fuel cost \$/h (System 2)

Method	Min	Max	Median	Average	Std
LWSO	111,497.63	111,497.63	111,497.63	111,497.63	0.000162
WSO	111,497.8	111,497.9	111,497.8	111,497.8	0.028971
GJO	111,498.5	111,501.6	111,499.5	111,499.8	0.978837
NGO	111,497.7	111,498.3	111,498.1	111,498	0.152862
FOX	111,498.2	112,266.3	111,518.6	111,618.3	223.1766
PSO	111,497.64	111,497.83	111,497.70	111,497.71	0.06644
SMA	111,497.65	111,509.36	111,497.78	111,499.72	3.43854
TSA	111,498.42	111,503.88	111,499.93	111,500.55	1.876489
JS	111,505.9	111,513.3	111,508.8	111,508.9	1.578434
HHO	111,503.9	111,664.6	111,565.6	111,566.9	35.06826
ISMA	111,497.63	111,497.72	111,497.65	111,497.66	0.02839

The best values obtained are in bold

Table 11 Optimal solution for the total generation cost of the third test system with 11-generator

Method	LWSO	WSO	GJO	NGO	FOX
P1 (MW)	57.11343	57.11342	57.19094	58.49754	57.42616
P2 (MW)	40.43608	40.43604	40.11951	39.5701	40.32525
P3 (MW)	57.88132	57.88106	57.51134	60.87681	58.45923
P4 (MW)	277.7031	277.7034	277.3188	275.3541	270.2858
P5 (MW)	186.8794	186.8792	189.4806	181.686	183.7433
P6 (MW)	249.186	249.1862	250.2199	249.0324	250.007
P7 (MW)	177.074	177.0738	174.615	178.1559	176.6359
P8 (MW)	380.1974	380.1976	382.8858	377.7396	385.7118
P9 (MW)	341.6148	341.6144	343.0202	344.6969	341.2841
P10 (MW)	378.5835	378.5834	374.0441	381.0812	385.1406
P11 (MW)	353.3309	353.3315	353.5938	353.3104	350.9809
V (MW)	4.55E-13	2.39E-10	5.2E-06	1.13E-03	3.70E-06
Fuel cost (\$/h)	12,274.4	12,274.4	12,274.45	12,274.55	12,274.57

The best values obtained are in bold

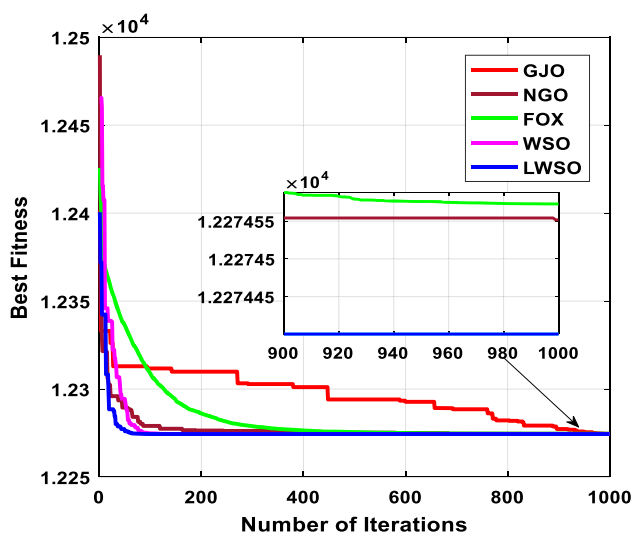
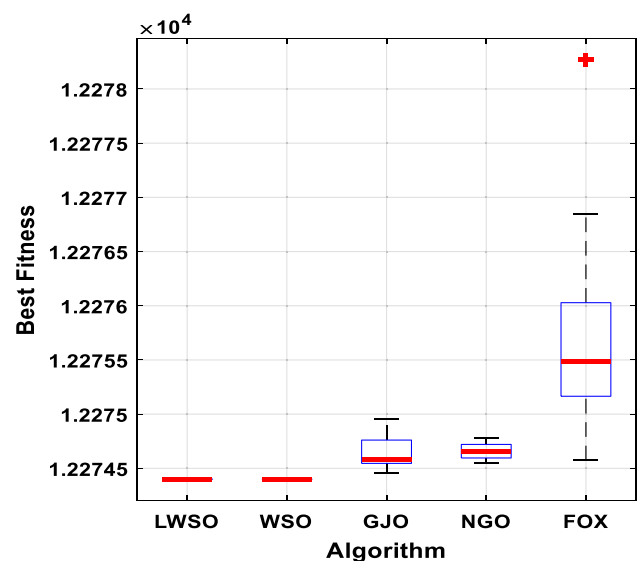
**Fig. 10** Fuel cost convergence curves of various algorithms (System 3)**Fig. 11** Boxplots of different algorithms (case study 3)

Table 12 Statistical results for fuel cost \$/h (System 3)

Method	Min	Max	Median	Average	Std
LWSO	12,274.4	12,274.4	12,274.4	12,274.4	3.73E-12
WSO	12,274.4	12,274.4	12,274.4	12,274.4	6.62E-11
GJO	12,274.45	12,274.96	12,274.58	12,274.65	0.154377
NGO	12,274.55	12,274.78	12,274.65	12,274.66	0.072292
FOX	12,274.57	12,278.27	12,275.48	12,275.69	0.869111
EBWO	12,274.46	12,274.7	12,274.54	12,274.55	0.065663
BWO	12,278.83	12,292.9	12,285.54	12,285.32	3.633401
SCSO	12,274.49	12,275	12,274.57	12,274.6	0.123994
SOA	12,282.33	12,305.01	12,289.87	12,291.6	6.814349
PSO	12,274.4	12,274.56	12,274.42	12,274.44	0.04804
SMA	12,274.4	12,274.41	12,274.41	12,274.41	0.001163
TSA	12,276.19	12,285.53	12,277.43	12,278.44	2.732958
JS	12,274.4	12,274.4	12,274.4	12,274.4	0.000652
HHO	12,274.4	12,274.45	12,274.41	12,274.42	0.0096
ISMA	12,274.4	12,274.41	12,274.4	12,274.4	0.001423

The best values obtained are in bold

laptop with a core i5 processor and 8 GB RAM. Table 6 records the parameters' settings of the tested algorithms. Four power systems with different generators and load levels are considered as: The first one has six generators and the load demand equals 283.4 MW; the second system involve 10 generators and the load demand increased to 2000 MW; and the third test system has 11 generators with loading level of 2500 MW. The last test system has 40 generators, and the loading level reaches 10,500 MW.

4.4.1 System 1: six thermal generators

Six generator units make up this system, and together they must provide 2.834 p.u. of power to meet the load requirement. The system characteristics include the cost coefficients and generation limits taken from [49]. Table 7 depicts the power outputs of each generator, total power loss, the total constraint violation (V), and total fuel cost associated with the six-unit power system and their comparison with numerous algorithms. It can be detected that LWSO algorithm gives the lowest value of total fuel cost compared with these recent techniques. The LWSO technique shows the effectiveness of objective function values and attains cogent and plausible results. Furthermore, the system constraints are satisfied.

The program implementation records the optimal fuel cost for each run, resulting in a collection of 20 optimal values per algorithm. These values are then used to generate boxplots, as shown in Fig. 6. A thorough analysis of these boxplots confirms the superior performance of LWSO technique compared to established optimization

techniques. Additionally, comparing the techniques' performances can be achieved by observing the fuel cost across iterations. Figure 7 presents the convergence curves of the techniques used in this test case. This figure approves the competence of the LWSO algorithm in reaching the lowest cost in minimum iterations. Furthermore, the comparison of the best, average, median, worst, and std values of the proposed LWSO and most recently published ELD algorithms such as ISMA [10], SMA [10], HHO [10], TSA [10], PSO [10], and JS [10] is tabulated in Table 8. It is clear from this table that the proposed LWSO technique succeeds in achieving the optimum solution for the ELD problem in this case.

4.4.2 System 2: ten thermal generators

In this system, the LWSO algorithm and the other techniques are verified on the ten-generator system. The population size and maximum iterations equal 1000 and 500, respectively. The generation bounds and coefficients of fuel cost were taken from the solution [10]. Table 9 shows the optimal fuel costs acquired using all techniques. It was proven that the LWSO technique achieves the best value of 111,497.63 \$/h, and the least violation 1.47E-06 MW compared with the original WSO, GJO, NGO, and FOX. The performance of the studied algorithms on the ELD problem is illustrated in Fig. 8, which depicts a boxplot. It is noteworthy that the LWSO algorithm has a narrower boxplot than the original WSO and other established algorithms in many instances, indicating greater consistency in terms of median, maximum, and minimum values. Figure 9 displays the convergence curves of the fuel cost. According to Fig. 9, the proposed LWSO algorithm converges to the minimum value in a number of iterations which is lower than the one for the original WSO technique. Furthermore, the comparison of the statistical measures' values of the proposed LWSO and most recently published ELD algorithms such as ISMA [10], SMA [10], HHO [10], TSA [10], PSO [10], and JS [10] is presented in Table 10. It is observable that the LWSO succeeds in achieving the optimum solution for this system.

4.4.3 System 3: eleven thermal generators

In case 3, the solution of ELD problem is obtained by the LWSO algorithm, and the load demand is PD = 2500 MW. The generation bounds and coefficients of fuel cost were taken from the solution [10]. The results acquired using the LWSO are compared with GJO, NGO, and FOX methods, as well as the original WSO algorithm which is shown in Table 11. These results show that the optimal solution found using the LWSO algorithm is less than the solution found using other well-known techniques. Figure 10

Table 13 The optimal solution for the fuel cost of the forty-unit system

Method	LWSO	WSO	GJO	NGO	FOX	Method	LWSO	WSO	GJO	NGO	FOX
P1	110.9654	111.1925	113.6833	110.8855	113.9983	P21	523.283	523.4478	525.4317	523.2815	523.3476
P2	110.9943	77.9684	112.6781	110.9345	113.9893	P22	523.305	523.4257	527.1042	523.2833	549.9879
P3	97.42296	97.5468	120	97.40107	120	P23	523.277	523.3181	526.4473	523.2952	532.1728
P4	179.7423	179.774	180.6674	179.7332	130.1123	P24	523.284	523.3792	530.0041	523.2861	523.3539
P5	88.36963	88.34238	87.88707	88.0054	97	P25	523.277	523.2416	536.0151	523.28	549.9994
P6	140	139.814	140	105.4092	140	P26	523.279	523.5083	526.7506	523.2889	523.7803
P7	259.6432	259.5433	276.916	259.6132	248.4128	P27	10.00	10.12625	14.3028	10.00492	10.0003
P8	284.6089	284.783	296.1365	284.6145	290.0736	P28	10.00	10.21457	10.84856	10.00333	10.00231
P9	284.6118	284.5785	287.3377	284.6104	299.77	P29	10.00	10.14601	10.18514	10.00159	10.02122
P10	130.0038	204.6317	131.344	130.0006	204.8026	P30	97	96.33838	89.92187	87.86835	97
P11	168.8049	243.5479	94	168.7994	168.7965	P31	190	181.105	190	159.915	189.999
P12	94.00018	168.8596	174.1461	168.8024	168.8108	P32	189.995	159.911	190	189.999	189.999
P13	214.7693	304.5492	129.2097	214.7632	304.5214	P33	189.997	163.521	189.992	189.999	190
P14	394.2811	394.2222	393.9505	394.2795	394.2786	P34	164.826	164.901	178.036	164.850	199.999
P15	394.2739	394.2383	394.326	394.2795	304.5188	P35	164.969	164.973	200	164.840	200
P16	394.2681	304.5736	304.7691	394.2786	394.2859	P36	165.02	167.813	175.599	164.8	200
P17	489.3012	399.7212	491.3892	489.2826	309.881	P37	109.999	91.0754	109.761	109.998	110
P18	489.308	489.2648	489.2901	489.2859	310.1625	P38	109.997	89.2555	107.652	109.996	109.999
P19	511.2919	511.2622	512.7854	511.2796	511.289	P39	89.2420	89.1315	96.4641	89.1723	109.999
P20	511.2867	511.3533	511.4638	511.2811	511.3617	P40	511.290	511.349	512.432	511.283	511.278
Objective function C(\$/h)			121,444.6	122,488.5	122,176.1		123,745.7				

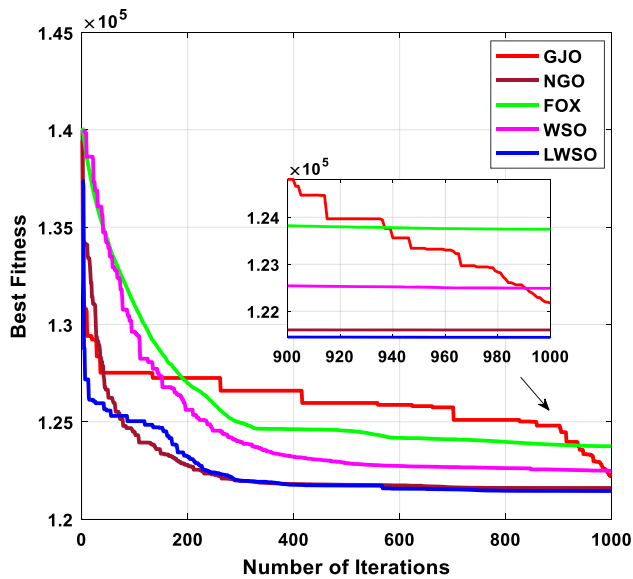


Fig. 12 Fuel cost convergence curves of various algorithms (System 4)

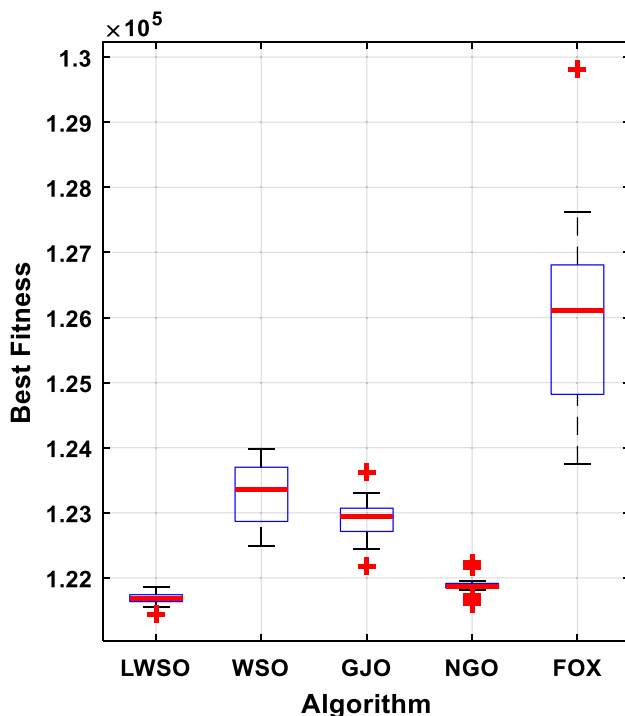


Fig. 13 Boxplots of various algorithms (System 4)

presents the convergence curve of the LWSO algorithm and other algorithms, which demonstrates the high accuracy and speed of the LWSO to reach the best solution compared to that of the studied intelligent techniques. The results of all algorithms for this case are represented in boxplots shown in Fig. 11 and show that the quartile of LWSO has a smaller range as compared to that of these recent techniques. Table 12 tabulates the best, worst,

average, median, and std values of various ELD solutions among 20 trial runs, which proves that LWSO succeeded in determining the optimal solution in comparison with other techniques including EBWO [45], BWO [45], SCSO [45], and SOA [45], ISMA [10], SMA [10], HHO [10], JS [10], TSA [10], and PSO [10].

4.4.4 System 4: forty generators

In the fourth system, assessments are executed on the forty-unit system. The generation bounds and coefficients of fuel cost were taken from [10]. The maximum iterations of the proposed LWSO, original WSO, and other studied techniques are increased to 1500 to increase the whole implementation for a large-scale power system. Table 13 shows the power schedules achieved using execution least economics. In this table, the fuel costs obtained by the LWSO algorithm are less than WSO, GJO, NGO, and FOX algorithms. The convergence curves of this case of the forty-unit system for reducing the total cost by the proposed LWSO and other studied techniques are displayed in Fig. 12. It is obvious that the performance of the LWSO for attaining the optimal solution to the economic load dispatch approves the strength of the LWSO algorithm. Figure 13 shows the boxplots of classification accuracy results achieved using the proposed LWSO and conventional WSO, GJO, NGO, and FOX algorithms for fuel cost. The boxplot represents the spread of classification precision values for each technique on a given dataset, with each technique being run twenty times. The boxplot enables comparison of techniques based on five essential statistical measures: minimum precision value, maximum precision value, lower quartile, upper quartile, and median. The higher the boxplot, the greater the classification accuracy achieved by the technique. The LWSO consistently delivers the highest classification accuracy values across most datasets. Figures demonstrate that the LWSO technique has a higher median and smaller interquartile ranges in most datasets, which serves as evidence of the technique's effectiveness and robustness.

Moreover, Table 14 shows the five essential statistical measures values of various ELD solutions via 20 individual runs. Additionally, Table 14 includes results from alternative techniques such as enhanced beluga whale optimization (EBWO) [45], beluga whale optimization (BWO) [45], sand cat swarm optimization (SCSO) [45], skill optimization algorithm (SOA) [45], ISMA [10], SMA [10], HHO [10], JS [10], TSA [10], PSO [10], parallel particle swarm optimization (PPSO) [16], slap swarm algorithm (SSA) [16], marine predator algorithm (MPA) [16], multigroup marine predator algorithm (MGMPA) [16], and hybrid slap swarm algorithm (HSSA) [2]. These algorithms show varying levels of performance in terms of fuel cost

Table 14 Statistical results for fuel cost \$/h (System 4)

Method	Min	Max	Median	Average	Std
LWSO	121,444.6	121,855.6	121,679.2	121,680.2	98.69825
WSO	122,488.5	123,982.5	123,359.4	123,286.7	490.7263
GJO	122,176.1	123,620	122,940.9	122,900.6	320.9004
NGO	121,602.1	122,235.9	121,872	121,872.8	149.3719
FOX	123,745.7	129,811.1	126,112.2	125,954.3	1391.512
EBWO	121,600.9	122,180.9	121,991.8	122,012.6	163.4211
BWO	122,875.8	123,858.9	123,395.8	123,398.6	240.7044
SCSO	123,633.3	128,464.7	125,210.6	125,219.7	1092.271
SOA	125,704.4	128,066.9	127,074.1	127,019.5	667.1643
PSO	121,627.99	122,077.55	121,892.71	121,893.32	151.4660
SMA	121,621.68	121,994.65	121,781.88	121,770.54	153.4794
TSA	125,385.34	126,380.35	125,368.53	125,628.15	661.7595
JS	122,577.7	123,413	123,331	123,181.8	246.9075
HHO	122,439.237	123,801.25	122,974.36	122,966.90	364.4271
ISMA	121,546.89	121,859.73	121,726.95	121,702.82	164.1745
PPSO	125,503.09	129,631.35	–	129,631.35	1033.37
SSA	123,565.75	127,442.23	–	127,442.23	905.64
MPA	123,180.98	126,614.40	–	126,614.40	927.37
MGMPA	122,634.69	125,523.19	–	125,523.19	755.10
HSSA	121,960.27	–	–	–	–

The best values obtained are in bold

Table 15 The value of the objective function for the studied cases using the proposed algorithms

Method	Case 1	Case 2	Case 3	Case 4
LWSO	605.9984	111,497.63	12,274.4	121,444.6
WSO	605.9984	111,497.8	12,274.4	122,488.5
GJO	606.0342	111,498.5	12,274.45	122,176.1
NGO	606.3391	111,497.7	12,274.55	121,602.1
FOX	606.041	111,498.2	12,274.57	123,745.7

The best values obtained are in bold

optimization. Overall, the statistical results in Table 14 affirm the effectiveness of the LWSO algorithm in achieving lower fuel costs compared to other techniques, showcasing its potential as a robust optimization approach for the forty-unit power system.

Table 17 Friedman test for the five algorithms

Function	LWSO	WSO	GJO	NGO	FOX
Case 1	1	2	3.9	4.95	3.15
Case 2	1	2.05	4.1	2.95	4.9
Case 3	1	2	3.4	3.7	4.9
Case 4	1.1	3.85	3.2	1.9	4.95
Mean ranks	1.025	2.475	3.65	3.375	4.475

Table 15 presents the value of the objective function for various cases using different algorithms, demonstrating the effectiveness of the proposed LWSO algorithm. In Case 1, the LWSO algorithm yields a highly competitive objective function value of 605.9984, identical to the WSO algorithm. However, as we move on to the more complex scenarios, such as Case 2, Case 3, and Case 4, LWSO

Table 16 Statistical results of the Wilcoxon rank-sum test

LWSO vs Function	WSO		GJO		NGO		FOX	
	P	Winner	P	Winner	P	Winner	P	Winner
Case 1	8.01E-09	+	8.01E-09	+	8.01E-09	+	8.01E-09	+
Case 2	6.8E-08	+	6.8E-08	+	6.8E-08	+	6.8E-08	+
Case 3	7.99E-09	+	8.01E-09	+	8.01E-09	+	8.01E-09	+
Case 4	6.67E-08	+	6.67E-08	+	5.83E-05	+	6.67E-08	+
WRST (+ / = / -)	4/0/0		4/0/0		4/0/0		4/0/0	

consistently outperforms other algorithms. Notably, in Case 2, it produces a significantly lower objective function value of 111,497.63 compared to the next best, WSO, with 111,497.8. These results highlight LWSO's exceptional ability to optimize and find solutions that outshine its counterparts, offering a promising solution for a wide range of real-world problems and optimization tasks.

4.5 Wilcoxon's rank test results

Table 16 displays the outcomes of the test conducted between LWSO and each technique, employing a significance level of $\alpha = 0.05$. Additionally, the table presents the statistical findings for LWSO across various scenarios, indicating its superior, comparable, or inferior performance relative to the comparison algorithm. LWSO demonstrates superiority over other comparative techniques in the statistical results for four instances of ELD optimization problems, underscoring its substantial dominance in most functions compared to alternative techniques. Hence, it can be inferred that the proposed LWSO technique showcases the most effective performance when juxtaposed with other algorithms.

4.6 Friedman's rank test results

In Table 17, the statistical results derived from Friedman tests are showcased. A smaller ranking value corresponds to a superior algorithmic performance. Based on the results, the rankings of the five algorithms are as follows: LWSO, WSO, NGO, GJO, and FOX. The highest rank signifies that LWSO stands out as the top-performing algorithm among the five considered.

5 Conclusion

In this article, an efficient LWSO algorithm has been proposed to attain the optimum solution to the ELD problem. Using the leader strategy in LWSO enhances the exploitation capability to avoid local optima and improve the local and global search. The performance of the LWSO algorithm has been assessed by using thirteen benchmark optimization problems that exist in the CEC2020 test suite, and it has been found that LWSO achieved better or similar results than DBO, FOX, and MFO algorithms and the conventional WSO. The efficacy of the LWSO has been assessed using evaluation metrics and statistical tests. Results from the Friedman ranking test and the Wilcoxon signed-rank test indicate a significant enhancement in the solution accuracy of the sizing problem when employing the LWSO, surpassing the performance of these recent algorithms. On another problem set, the proposed LWSO

algorithm is verified on 6-, 10-, 11-, and 40-thermal units-based test systems. Furthermore, the performance of LWSO in addressing the ELD problem has been scrutinized, revealing a consistently superior success rate compared to WSO, GJO, NGO, and FOX algorithms. The statistical analysis based on 20 individual runs has been carried out for each case, and the results are compared with the previous works of the literature. The dominance of the LWSO algorithm over others has been confirmed. In the future work, the LWSO algorithm could be applied to effectively solve other complex optimization problems in several fields.

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Data availability All required data are involved in the text.

Declarations

Conflict of interest Authors have no conflict of interest.

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